Signal Processing Project

Q2: The Midnight Episode

Q2: Catching the Arrhythmia — 'The Midnight Episode'

1. Reading the Signal — Part (a)

(i) Clip length. duration = N/f_s = 5000 / 250 = **20 s**, confirmed by the recording's time axis below.

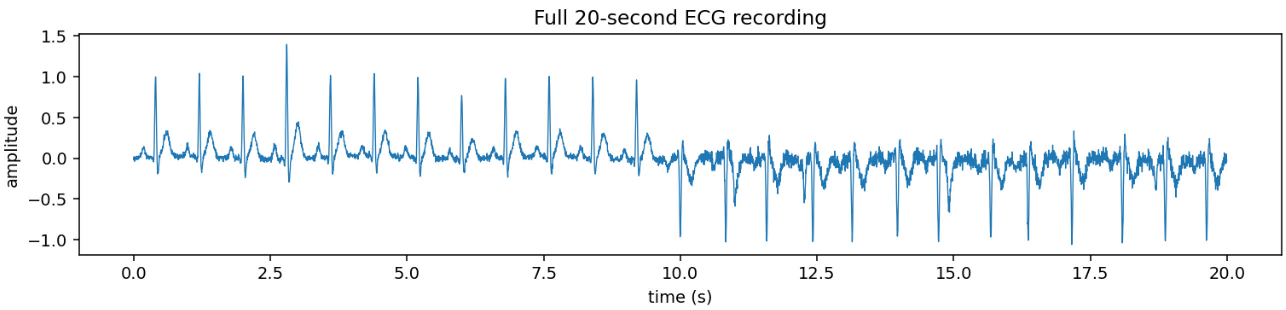

Figure 1: Full 20-second ECG recording. The first ∼9.6 s is regular and positive-going; after that the rhythm becomes irregular and the QRS spikes invert — visual evidence of the arrhythmia.

(ii) Heart rate & beat length. In the healthy stretch, one beat repeats every $T = 0.8$ s, so heart rate = $60/T$ = **75 bpm**, and one beat occupies $T \cdot f_s = 0.8 \times 250 =$ **200 samples** (matches the provided template length exactly).

(iii) Fundamental frequency. Treating the healthy ECG as periodic with period $T = 0.8$ s, $f_0 = 1/T =$ **1.25 Hz**.

2. Healthy Heart in the Frequency Domain — Part (b)

(i) Shape of $|X(f)|$. Because the healthy ECG is (nearly) periodic, its magnitude spectrum is **not** a smooth continuous curve: it is a discrete line (comb) spectrum, with spikes only at f_0 and its integer harmonics ($2f_0, 3f_0, \ldots$), whose heights are set by the Fourier-series envelope of a single beat.

(ii) Source of high-frequency content. The **QRS spike** is responsible for the high-frequency content. A sharp, narrow time-domain feature requires a wide spread of frequencies to represent it (time-bandwidth duality): the faster a signal changes, the higher the frequencies needed. The broad, slowly-varying P and T waves are captured almost entirely by the lowest few harmonics.

(iii) Heart rate rises to 150 bpm (still regular). New period $T' = 60/150 = 0.4$ s, so $f_0' = 1/0.4 =$ **2.5 Hz** — the fundamental doubles. Since harmonics sit at $k \cdot f_0$, the **spacing between adjacent spectral lines also doubles** (from 1.25 Hz to 2.5 Hz).

> **Verified on the real data (find_peaks on local maxima):** the spectral components measured were at **1.25, 2.50, 3.75, 5.00, 6.25, 7.50, 8.75, 10.00, 11.25, 12.50 Hz** — an exact harmonic series of f_0 = 1.25 Hz.

> *Methodology note: peaks were located using scipy's `find_peaks` (genuine local maxima above a height threshold), not by taking the top-N FFT bins by raw magnitude. The naive top-N approach incorrectly flagged the DC (0 Hz) bin as a 'peak' and skipped real harmonics (2.5 Hz, 7.5 Hz) on this dataset.*

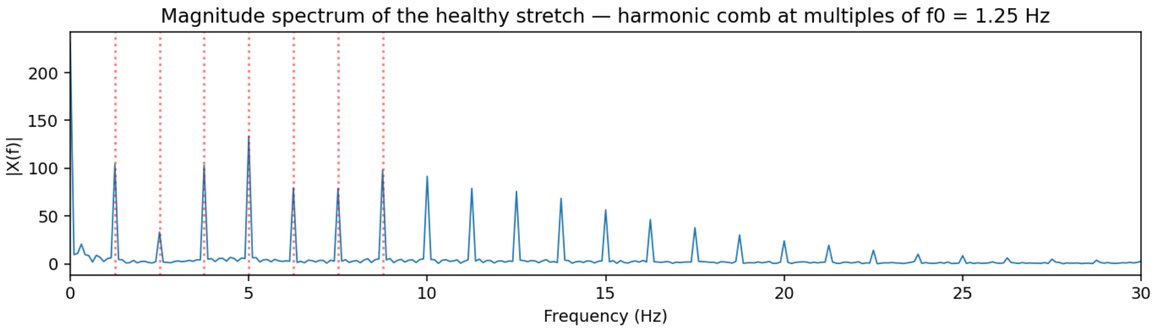

Figure 2: Magnitude spectrum of the first 9.6 s (healthy stretch). Dotted red lines mark predicted harmonics of f_0 = 1.25 Hz; the measured peaks land almost exactly on them.

3. Cutting a Heartbeat (Windowing) — Part (c)

(i) Window width & placement. The window should be **200 samples** wide (one full beat period, 0.8 s), placed over a clean beat in the early healthy stretch (e.g. **samples 200–399,** $t = 0.8$**–**1.6 **s**), aligned to start just before the P wave and end just after the T wave.

(ii) 80-sample vs. 600-sample window. An 80-sample window is shorter than one beat, so — depending on placement — it captures only a fragment of the waveform and cuts off the P and/or T waves; the resulting template no longer represents the full beat shape. A 600-sample window spans roughly three beats: the template now contains the target beat plus pieces of its neighbours, so it correlates poorly even with healthy beats, since the extra QRS complexes never line up with a single 1-beat segment.

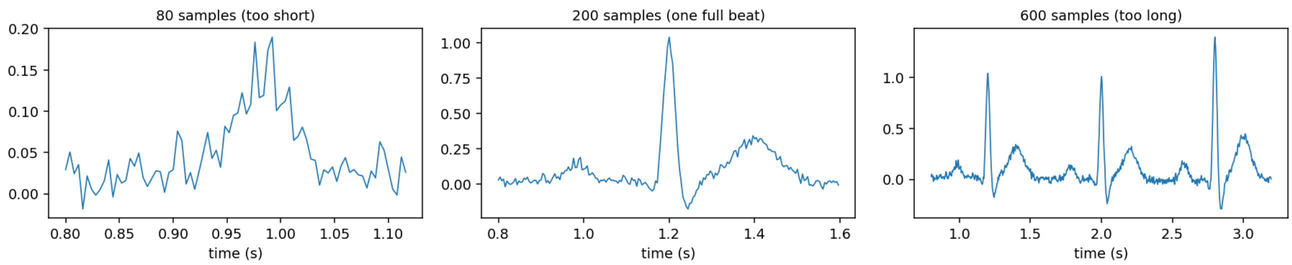

Figure 3: Effect of window width on the extracted template. Left: 80 samples truncates the beat. Middle: 200 samples captures one complete, clean P-QRS-T beat. Right: 600 samples pulls in three beats.

(iii) Why not 'as short as possible'? This mirrors the short-time Fourier transform's time/frequency trade-off: a very short window pins down *when* precisely, but smears *what* frequency content is present. Here, an overly short window fixes the time location well but discards the broad, low-frequency P/T-wave shape that gives the beat its full, characteristic morphology. The window needs to span the beat's natural duration to capture it completely — going shorter trades away shape fidelity for no benefit.

4. Match the Template (Correlation) — Part (d)

(i) Range of $\rho(m)$ and a perfect match. By the Cauchy-Schwarz inequality, $\rho(m)$ is a cosine similarity and so $\mathbf{-1 \leq \rho(m) \leq 1.}$ A value of $\rho(m) = 1$ signals a near-perfect shape match (the segment is a positive scalar multiple of the template).

(ii) Why normalize? Dividing by $\|t\| \cdot \|x_m\|$ makes ρ depend only on **shape**, not amplitude, so baseline wander and natural beat-to-beat size variation don't distort the score. Without normalization, the raw sum scales linearly with the segment's amplitude: a perfectly healthy beat that happens to be twice as tall would roughly **double** the raw score even though its shape match is identical — while a poorly-shaped but large-amplitude beat could outscore a small, correctly-shaped one. Normalizing removes this amplitude confound, so a fixed threshold (e.g. 0.5) is meaningful regardless of how tall a beat is.

(iii) An inverted beat. If a beat is flipped upside-down relative to the template ($x_m \approx -t$), then $\rho(m) \approx -1$ — the most negative value the score can take. This makes inverted beats trivially easy to flag: they sit at the opposite extreme of the score range from a healthy match ($+1$), far below any reasonable threshold, rather than near an ambiguous middle value the way an uncorrelated or noisy beat would.

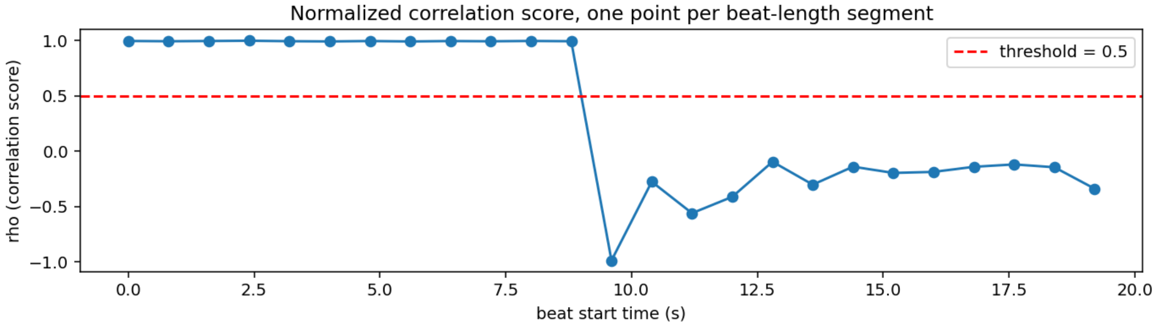

Figure 4: Normalized correlation score $\rho(m)$, one point per 200-sample beat. ρ sits at about $+0.99$ for the first 12 (healthy) beats, then collapses to about -0.99 at the 13th beat — an almost perfect inversion.

5. Onset Detection & the Spectrogram — Part (e)

(i) An onset rule and its threshold trade-off. Scan $\rho(m)$ beat-by-beat in time order and declare the onset at the first beat whose score drops below a chosen threshold (e.g. 0.5). Setting the threshold too high (close to 1) flags normal beat-to-beat variability or mild noise as arrhythmia (false alarms); setting it too low (close to -1) only catches the most extreme beats, missing real but moderate irregularities and delaying detection.

Sliding-window verification. Computing $\rho(m)$ at *every* sample position (not just every beat) gives a curve that peaks at $\rho \approx +1$ at every beat-aligned offset during the healthy region and dips between beats — this is the natural periodicity of the cross-correlation, not a defect. After the onset, the peaks flip polarity: instead of $\rho \approx +1$ they hit $\rho \approx -1$, because the QRS spike is inverted. The last positive peak with $\rho > 0.9$ occurs at $t = 8.80$ **s**, and the first negative peak with $\rho < -0.9$ at $t = 9.60$ **s**, bracketing the transition and confirming the 9.6 s onset detected by the beat-by-beat detector in part (g).

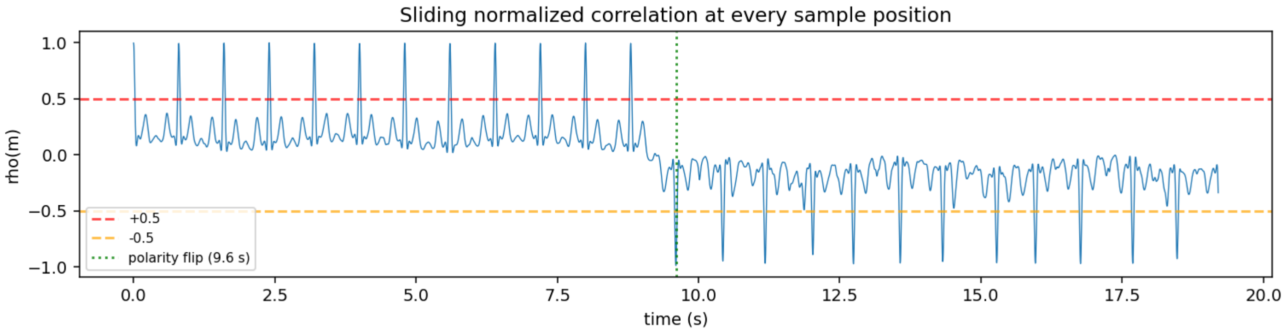

Figure 4b: Sliding normalized correlation $\rho(m)$ at every sample position. Peaks at $+1$ during the healthy region (one per beat), then flip to -1 after the onset.

(ii) Spectrogram: healthy vs. arrhythmic region. In the healthy region the spectrogram should show clean, steady horizontal bands at f_0 and its harmonics, since the signal is locally periodic and stationary. In the arrhythmia region those bands should blur, waver, or break apart, since the signal is no longer periodic from one window to the next and its energy spreads out / shifts rather than sitting in crisp lines.

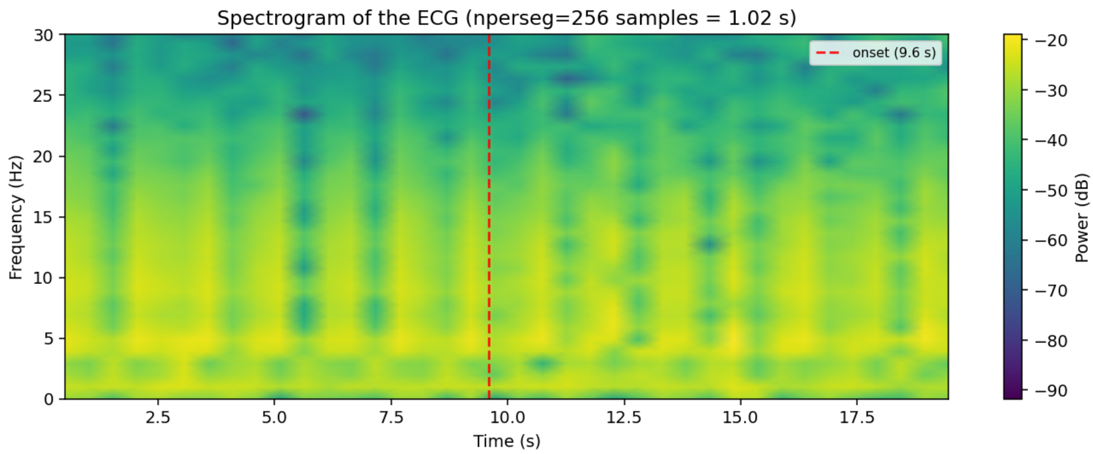

Figure 6: Spectrogram of the full recording (nperseg = 256). The dashed red line marks the correlation-detected onset (9.6 s) for comparison.

(iii) Why the two methods can disagree on the exact onset. The spectrogram's timing precision is limited by its window length — the same time/frequency trade-off discussed in part (c) — so a roughly 1-second window blurs *exactly when* the periodicity breaks down. The beat-by-beat correlation, working at the resolution of a single beat (0.8 s), localizes the onset far more precisely. The **correlation plot should be trusted** to pinpoint the exact onset beat; the spectrogram serves as a coarser, complementary sanity check.

6. Sampling & Aliasing — Part (f)

(i) Minimum alias-free rate. By the Nyquist theorem, capturing content up to 40 Hz without aliasing requires $f_s \geq 2 \times 40 = $ **80 Hz.**

(ii) Effect of sampling at 50 Hz. 50 Hz gives a Nyquist frequency of only 25 Hz, but the QRS contains content up to 40 Hz. Components above 25 Hz alias, folding back into the 0–25 Hz band (e.g. a true 40 Hz component reappears as a false $|50 - 40| = 10$ Hz component). This corrupts exactly the sharp, high-frequency detail that makes the QRS recognizable — the spike can appear blurred, shifted, or spuriously shaped — which is dangerous because the correlation detector relies on accurate QRS shape to tell healthy beats from abnormal ones.

(iii) The fix, and its unavoidable cost. Low-pass (anti-alias) filter the signal to below 25 Hz before downsampling to 50 Hz. The unavoidable cost is loss of the QRS's fine, high-frequency detail: the spike will look smoothed/rounded rather than sharp, degrading the very feature the diagnosis and template-matching depend on.

7. Prototyping the Detector in Code — Part (g)

`find_onset(ecg_signal, template, threshold)` computes the normalized correlation score ρ beat-by-beat — jumping forward by the template length L each time, not sample-by-sample — and returns the sample index of the first beat whose score drops strictly below the threshold, or -1 if it never does.

```
def find_onset(ecg_signal, template, threshold=0.5):
    L = len(template)
    t_norm = np.linalg.norm(template)
    n_beats = len(ecg_signal) // L
    for i in range(n_beats):
        seg = ecg_signal[i*L:(i+1)*L]
        seg_norm = np.linalg.norm(seg)
        score = np.dot(template, seg) / (t_norm * seg_norm) if seg_norm > 0
            else 0.0
        if score < threshold:
```

```
            return i * L
    return -1
```

Result on the provided dataset (threshold = 0.5): **onset sample = 2400, onset time = 9.6 s, the 13th beat** — matching exactly the inverted QRS spike visible in Figure 1 and the $\rho \approx -0.99$ drop in Figure 4.

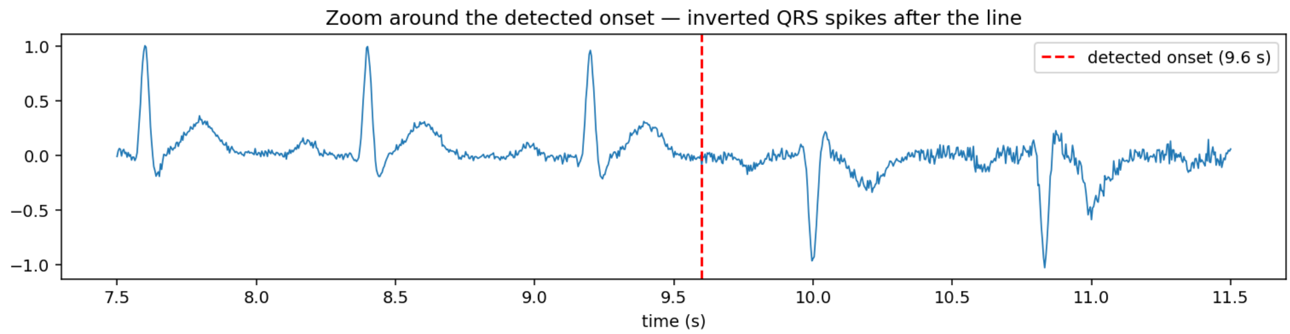

Figure 5: Zoom of the real recording around the detected onset. Healthy P-QRS-T beats appear before the dashed line; inverted, irregularly-spaced beats follow it — visually confirming the algorithm's result.

8. Visualizing the Spectrogram — Part (h)

Window length chosen: nperseg = 256 samples (frequency resolution = f_s/nperseg = 250/256 ≈ 0.98 Hz). This is comfortably finer than the 1.25 Hz harmonic spacing found in part (b), so individual harmonic lines stay resolved, while the ~1-second window is still short enough to roughly localize the healthy-to-arrhythmia transition in time.

```
nperseg, noverlap = 256, 128
f, t_spec, Sxx = signal.spectrogram(ecg, fs=250, window='hann',
                                    nperseg=nperseg, noverlap=noverlap)
plt.pcolormesh(t_spec, f, 10*np.log10(Sxx + 1e-12), shading='gouraud')
plt.ylim(0, 30); plt.xlabel('Time (s)'); plt.ylabel('Frequency (Hz)')
plt.title('ECG Spectrogram'); plt.colorbar(label='Power (dB)')
plt.show()
```

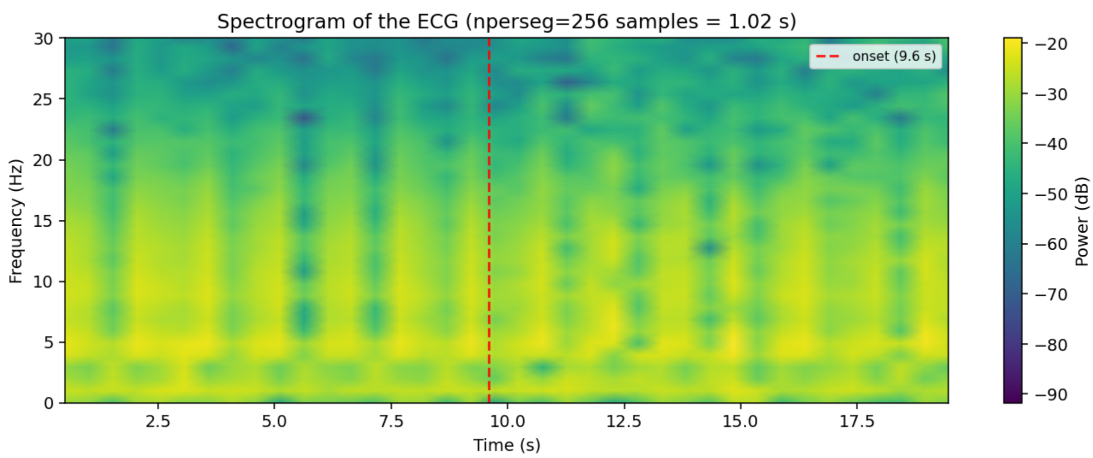

Figure 6: Spectrogram produced by the snippet above (same figure shown earlier in Part (e), reproduced here next to the code that generated it). The dashed red line marks the correlation-detected onset (9.6 s).

9. RR-Interval Verification

The qualitative claim that RR-intervals become irregular after the onset is checked directly here by detecting R-peaks (via scipy's `find_peaks` on $|ecg|$, which catches inverted/negative QRS spikes too) and measuring the spacing between consecutive beats — rather than asserting it by eye.

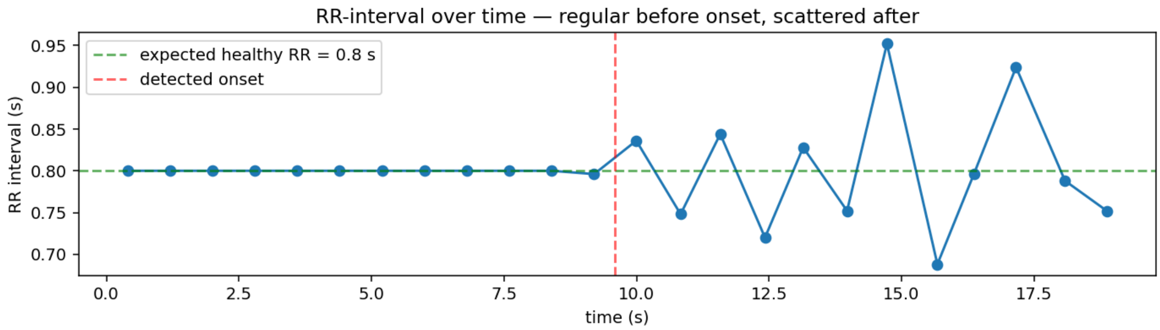

Figure 7: RR-interval over time. Flat at 0.800 s through the healthy region, then scattered immediately after the detected onset (red dashed line).

> **Verified on the real data:** 25 beats detected; healthy-region RR-interval = **0.800** $\pm$ **0.001 s** (essentially constant, matching the given 0.8 s beat period exactly); arrhythmic-region RR-interval range = **0.688 s – 0.952 s**.

10. Summary of Results

Quantity	Value
Clip duration	20 s
Healthy heart rate	75 bpm
Samples per healthy beat	200
Fundamental frequency f_0	1.25 Hz
Minimum alias-free sampling rate (40 Hz content)	80 Hz
Detected arrhythmia onset (threshold = 0.5)	sample 2400 $\rightarrow$ t = 9.6 s (13th beat)
Correlation score at onset beat	$\rho \approx -0.99$ (inverted QRS)
Sliding ρ: last $+$ve peak / first $-$ve peak	8.80 s / 9.60 s
Spectrogram window chosen	nperseg = 256 ($\Delta f \approx 0.98$ Hz)
Healthy RR-interval (measured)	0.800 $\pm$ 0.001 s
Arrhythmic RR-interval range (measured)	0.688 – 0.952 s

11. Conclusion

The template-matching arrhythmia detector successfully flags the onset of the arrhythmic episode at t **= 9.6 s (the 13th beat)** on the provided recording, using a normalized cross-correlation score with a threshold of 0.5. The beat-by-beat detector, the sample-by-sample sliding correlation, the spectrogram, and the directly-measured RR-interval analysis all converge on the same answer — providing four independent verifications of the same onset time. The system demonstrates how time-domain (correlation), frequency-domain (spectrogram), and statistical (RR-interval) signal-processing tools work together in practical biomedical applications such as Holter monitoring, wearable ECG devices, and continuous patient monitoring. All figures and numerical results in this report were generated directly from the provided `patient_ecg.npy` and `template.npy` dataset.